

CM5 socket

File: cm5_socket.kicad_sch

System MCU, Watchdog, EEPROM

File: system_mcu.kicad_sch

USB3 Hub, Front USB-C

File: usb3_hub.kicad_sch

DIB 2.0, TFT DSI

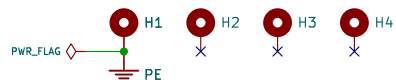
File: dib.kicad_sch

Miscellaneous

File: misc.kicad_sch

Ext video, Ethernet, Rear USB

File: ext_video.kicad_sch



<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

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Sheet: /

File: BB3plus test mainboard.kicad_sch

Title: BB3+ test mainboard

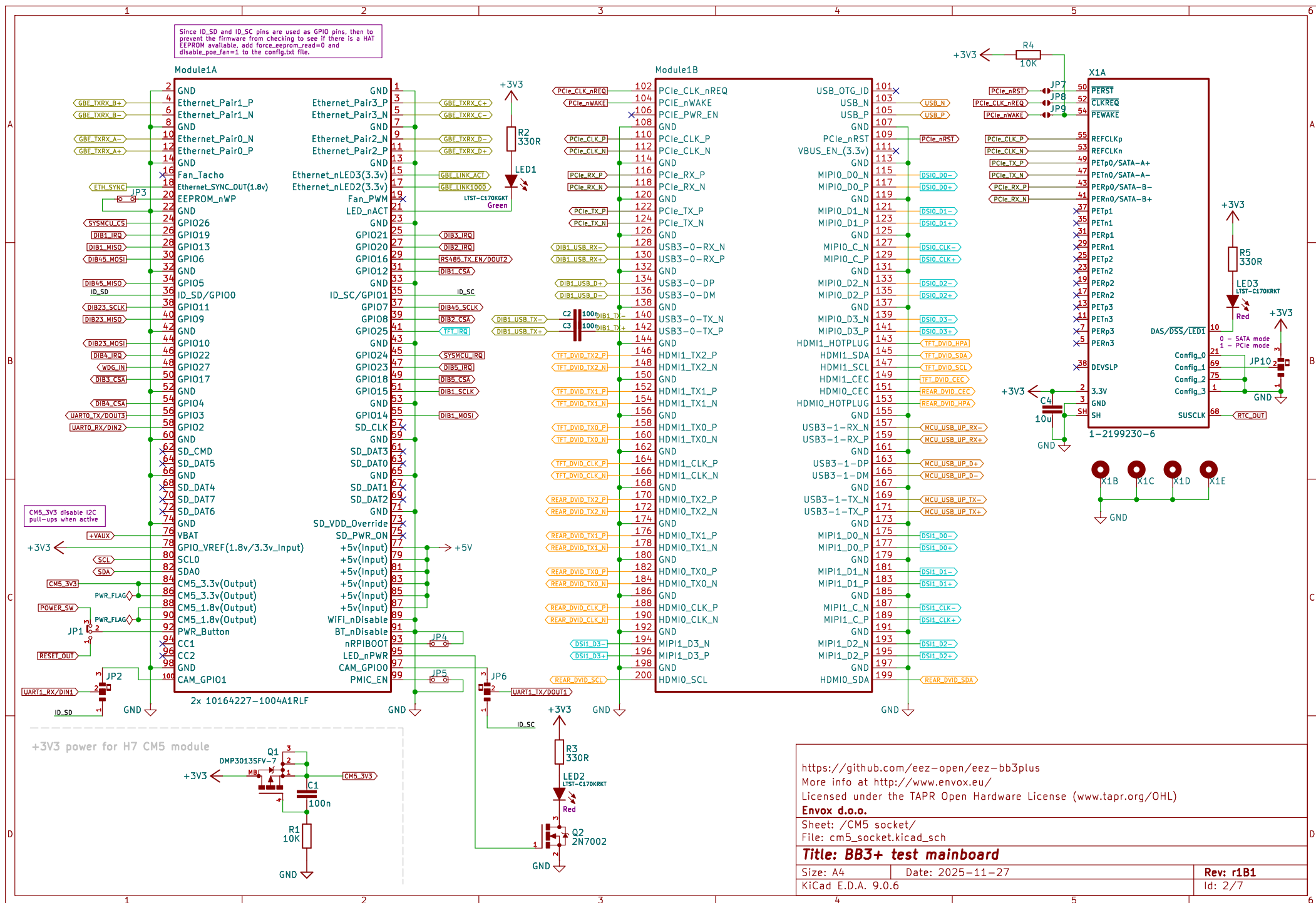
Size: A4

Date: 2025-11-27

Rev: r1B1

Size: A4	
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Id: 1/7



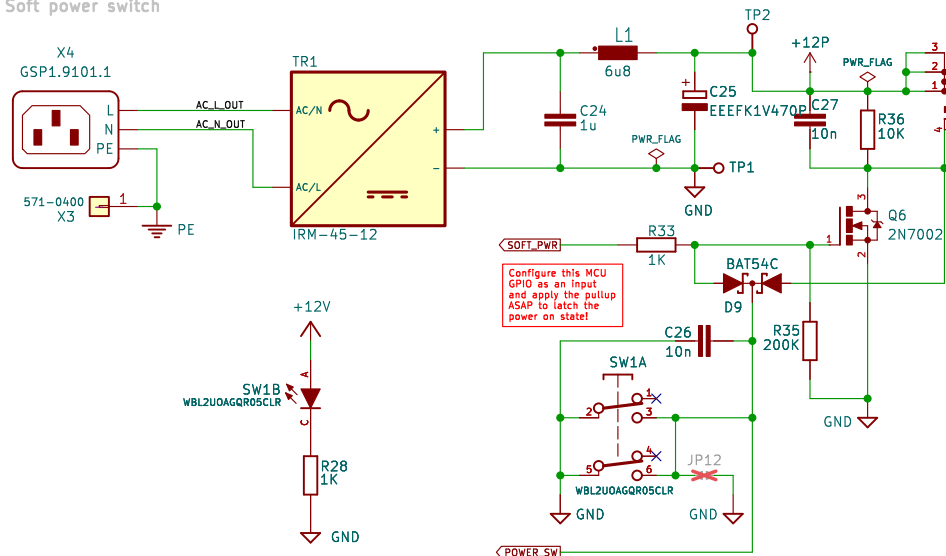
The System MCU is programmed so that the Master MCU first simultaneously activates DIB4_CSA and DIB5_CSA and then simultaneously activates DIB2_CSA and DIB3_CSA which will reset the System MCU and put it into bootloader mode. Simultaneous activation of two CSA (Chip Select) signals on the same SPI channel should never occur in normal operation. 100K pullup resistors ensure that RESET and BOOT0 are inactive if the Master MCU is not installed.



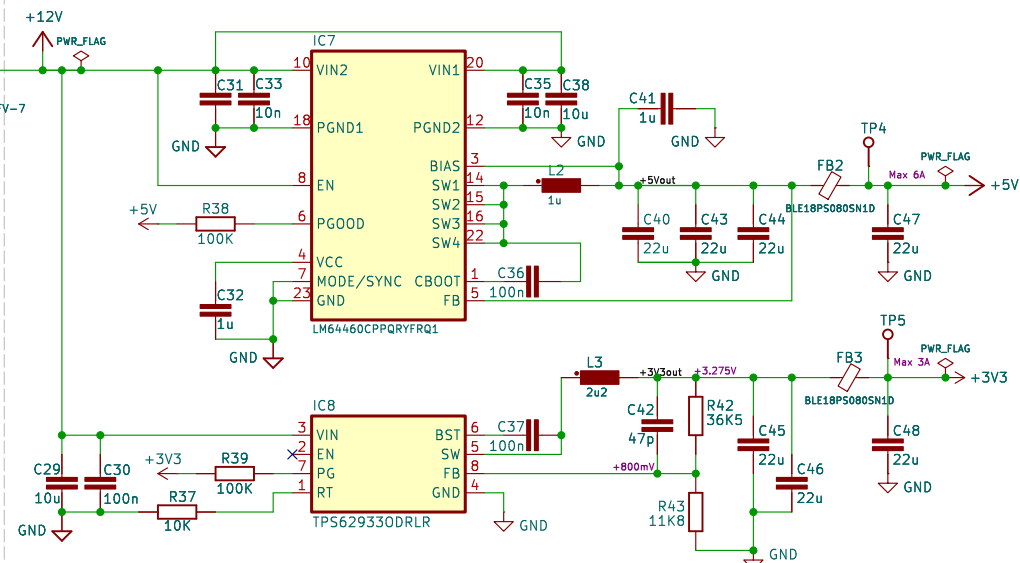
The watchdog signal is daisy chained in such a way that the master MCU/CPU sends a "heartbeat" signal WDG_IN to the system MCU (G071) which forwards it as WDG_OUT. The start of monitoring the "heartbeat" signal is controlled with WDG_SET.



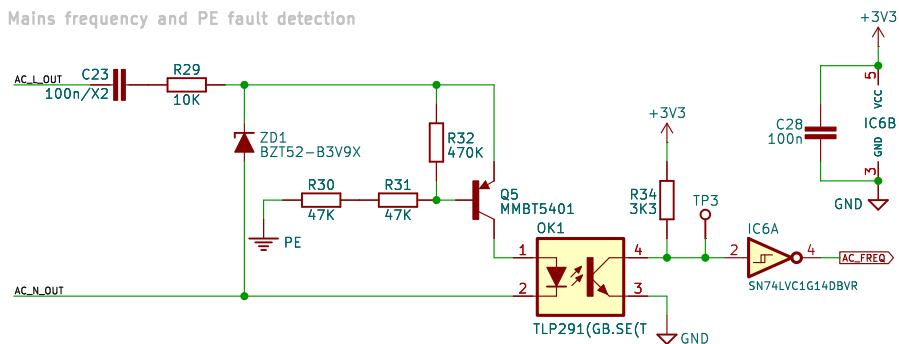
Soft power switch



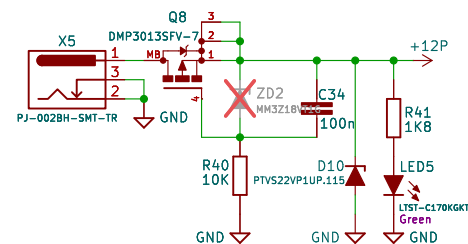
DC-DC stepdown converters (+5V, +3V3)



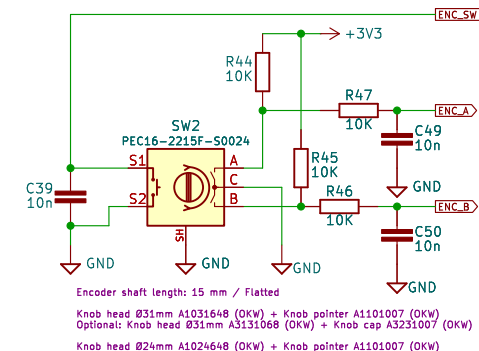
AC Mains frequency and PE fault detection



12Vdc aux



Encoder with switch



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Sheet: /Miscellaneous/

File: misc.kicad_sch

Title: BB3+ test mainboard

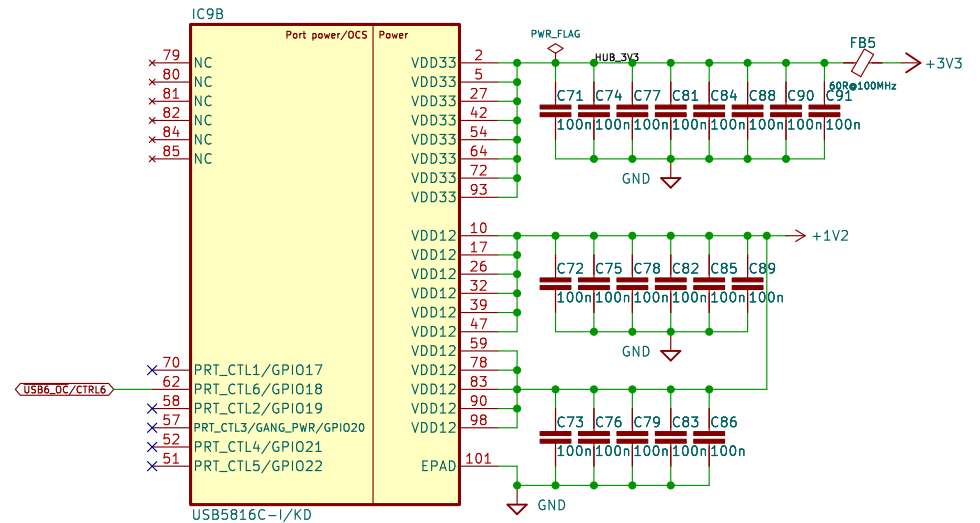
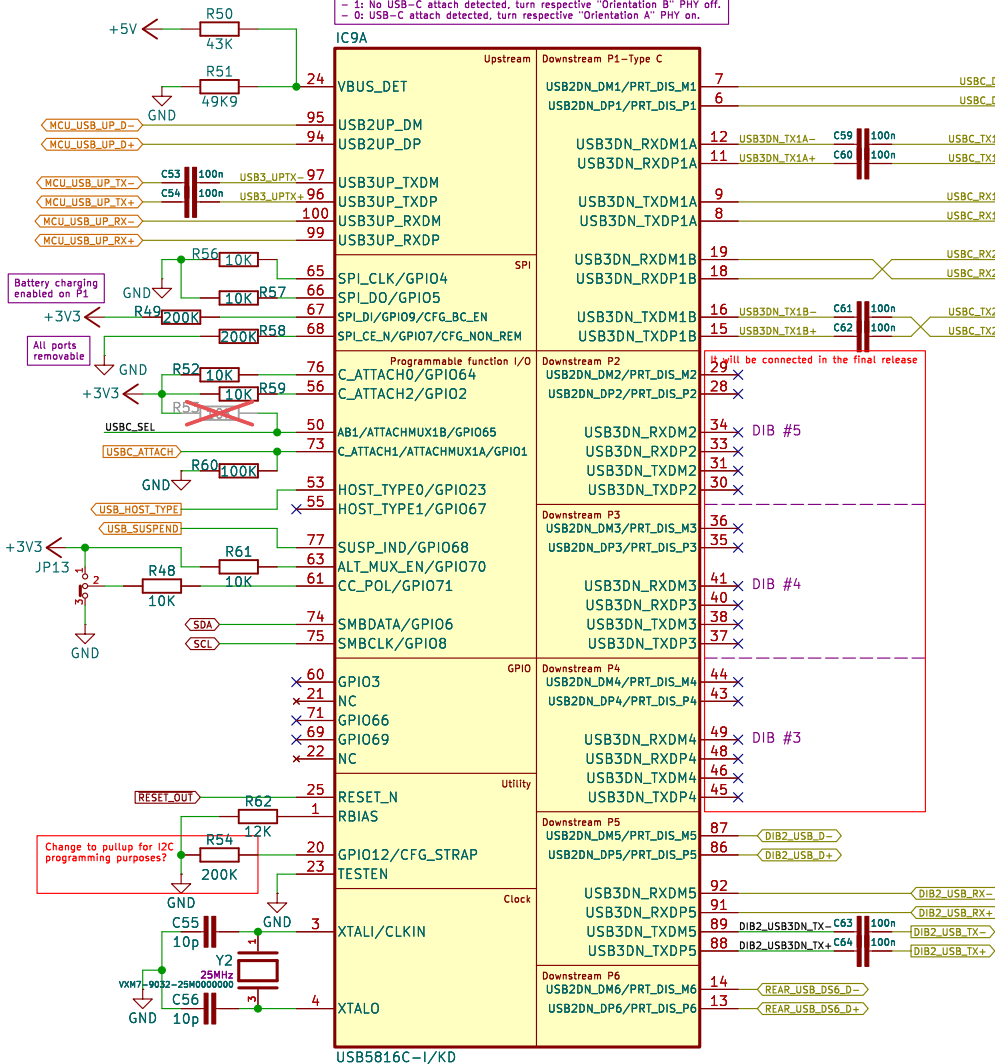
Size: A4 Date: 2025-11-27

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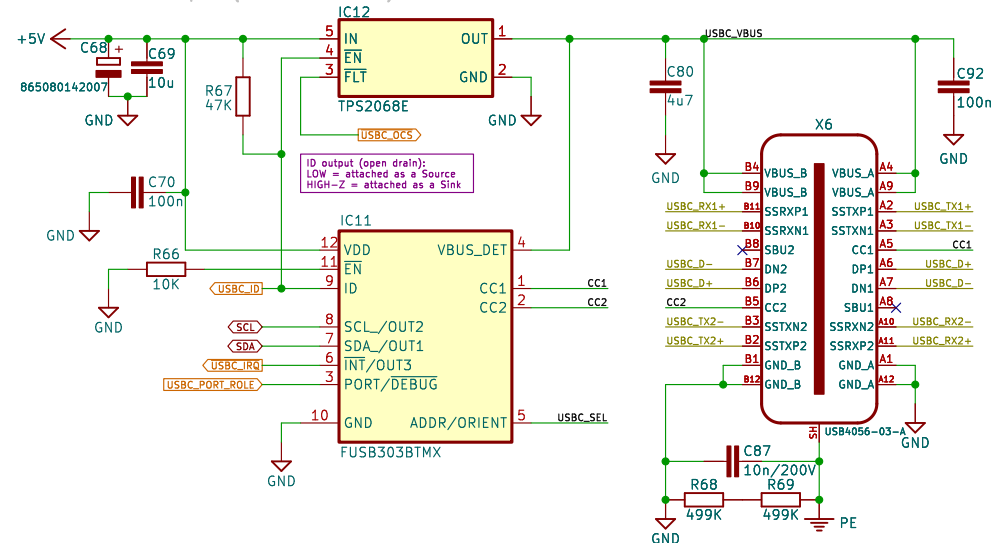
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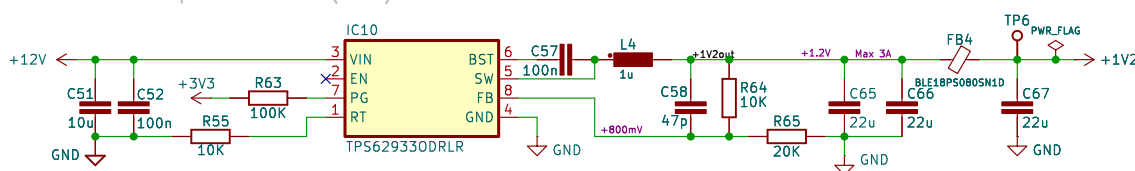
6-port USB hub



USB-C 3.0 front port (source max 1.5A)



USB hub DC-DC stepdown converter (+1V2)



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Sheet: /USB3 Hub, Front USB-C/

File: usb3_hub.kicad_sch

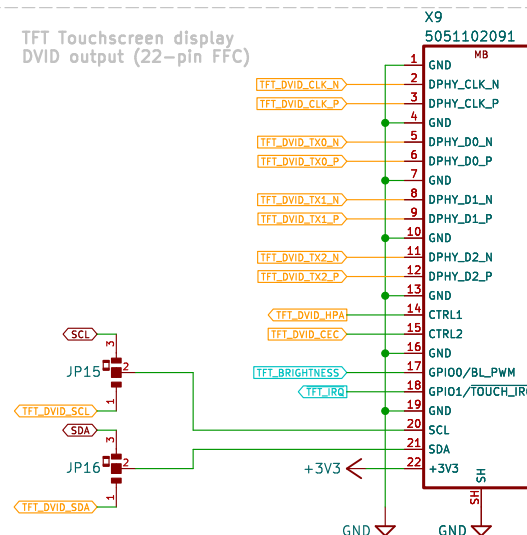
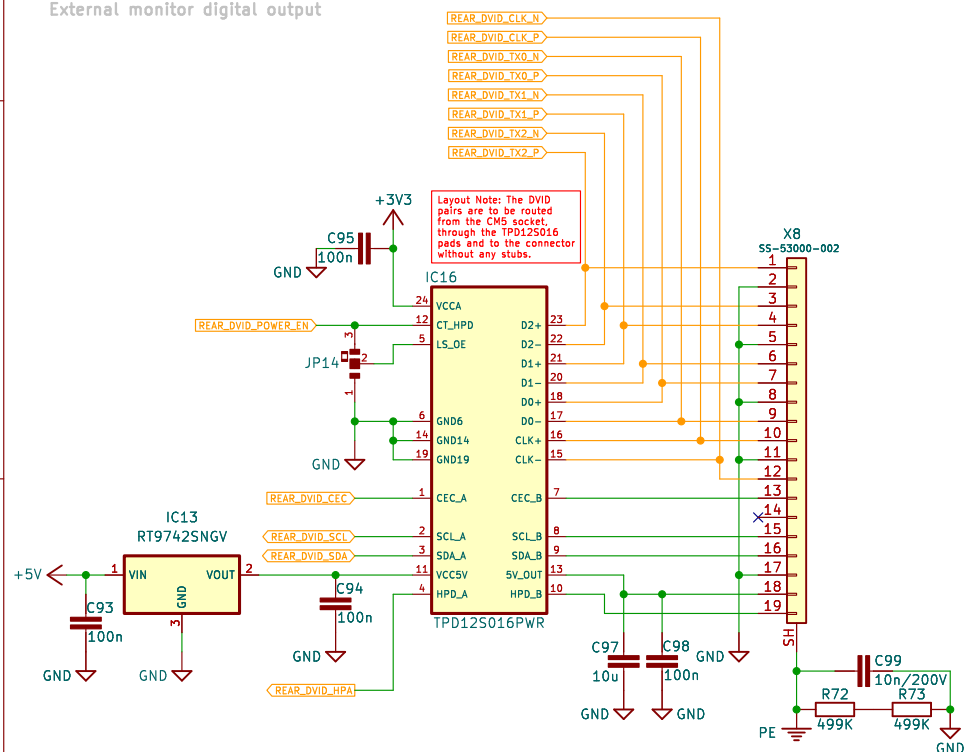
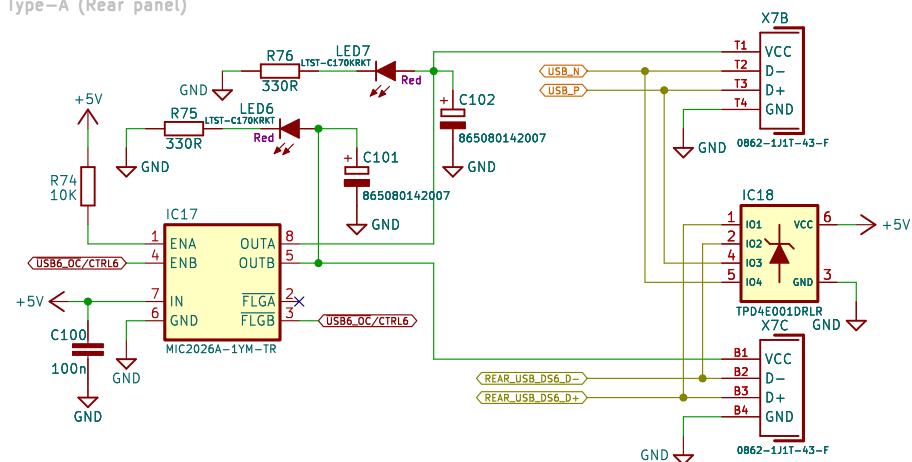
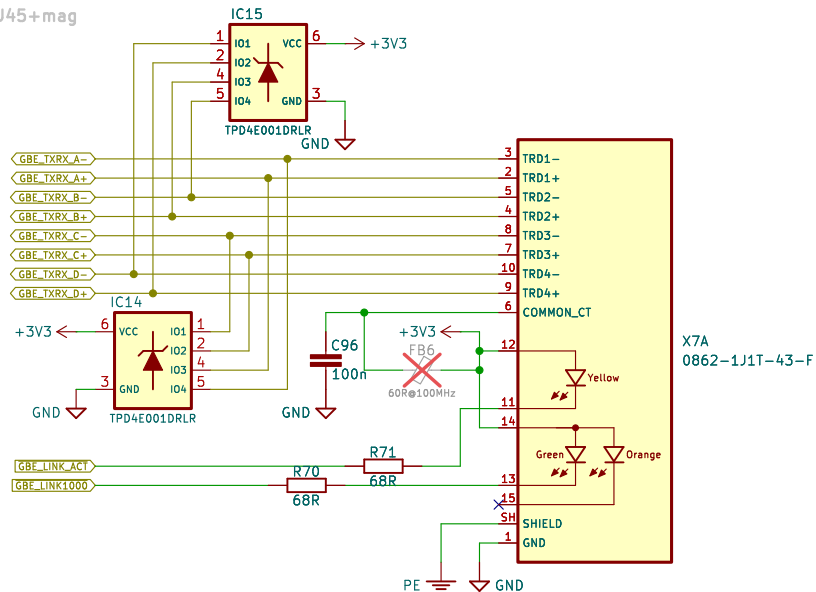
Title: BB3+ test mainboard

Size: A4 Date: 2025-11-27

KiCad E.D.A. 9.0.6

Rev: r1B1

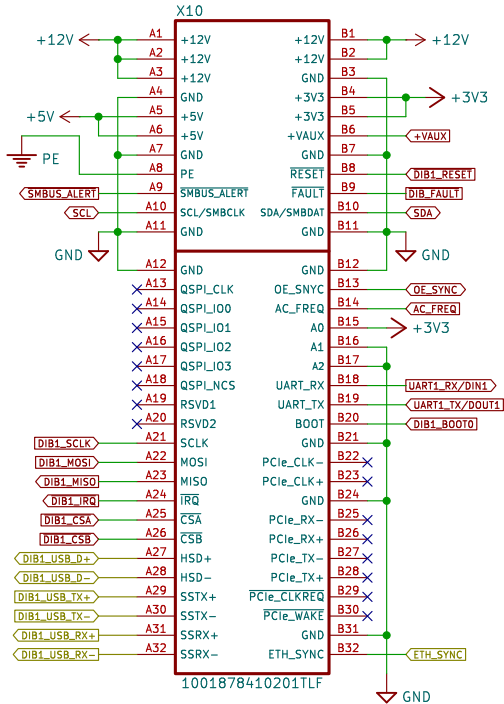
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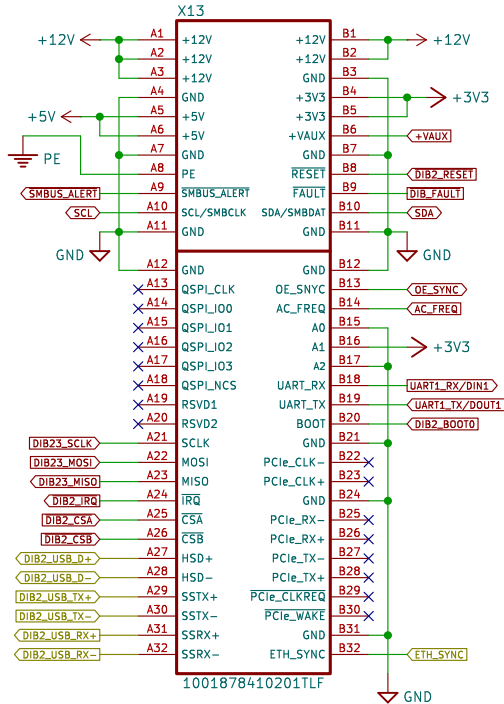
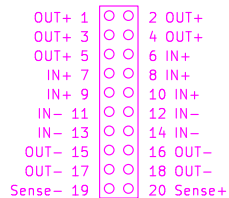
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KiCad E.D.A. 9.0.6

Id: 6/7

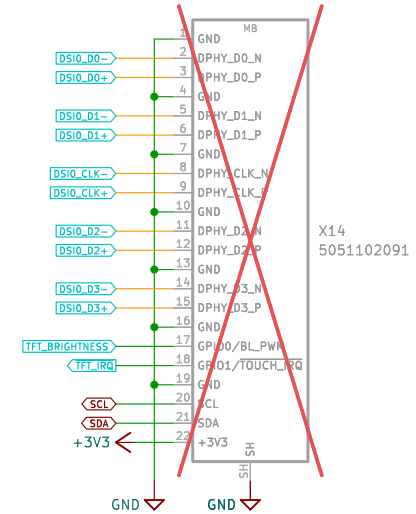
DIB 2.0 Signal connectors



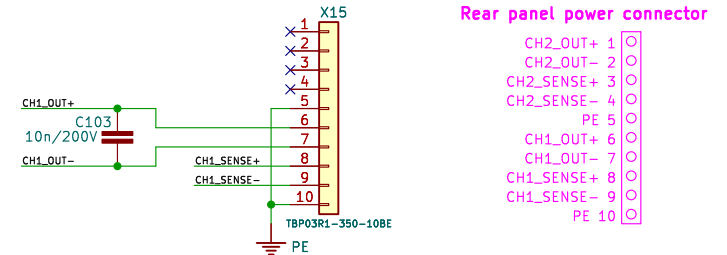
20-pin DIB v2.0 Power connector



TFT Touchscreen display DSI output (22-pin FFC)



Rear front power outputs



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